Nirusma Dahal

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Education

University of Illinois Urbana-Champaign — Champaign, IL
Master of Computer Science — *Aug. 2026 – Present*

University of Virginia — Charlottesville, VA
Bachelor of Arts in Computer Science, Minor in Data Science — *Aug. 2022 – May 2026*

Experience

Software Engineering Intern — June 2025 – Aug. 2025
Walmart Global Tech — *Sunnyvale, CA*

- Refactored legacy Spring Boot services by removing deprecated CCM logic and updating downstream workflows to preserve consistent behavior across interdependent components during a live migration.
- Validated system behavior by analyzing structured API logs and order-level responses, performing mock-to-production comparisons to detect schema and field-level inconsistencies prior to deployment.
- Migrated and updated 120+ integration test cases, isolating migration-related failures from pre-existing defects to improve debugging accuracy during rollout.
- Executed end-to-end API checks using Bruno by comparing order-based request and response payloads across environments to ensure parity between mock and production systems.

Data Engineering Intern — May 2024 – Aug. 2024
Amida Technology Solutions — *Washington, DC*

- Built a Python and PostgreSQL ETL pipeline to generate synthetic healthcare datasets from the MIMIC dataset, authoring SQL queries to extract and transform source records and validate output distributions, scaling to 16,000+ records across 26+ variables.
- Applied statistical validation techniques including mean/variance matching and KL divergence to ensure fidelity between synthetic and real patient distributions.
- Designed modular data generation workflows to support repeatable experimentation and reduce manual dataset preparation across research iterations.

Projects

Civic Radar
Next.js, TypeScript, Tailwind CSS, PostgreSQL, pgvector, OpenAI Embeddings, Claude API, Vercel

- Built and deployed a full-stack web app that matches Virginia legislation to a user's topics and county, ingesting bills from the Open States API and ranking them with OpenAI text embeddings and pgvector cosine similarity.
- Found that similarity scores alone could not separate relevant from irrelevant bills and added a cached Claude relevance-filtering stage, then generated plain-language bill summaries from the official bill text.
- Grounded Claude explanations in the bill text by requiring a supporting quote and verifying it against the source so unsupported claims are never shown; added anonymous thumbs up/down feedback to measure relevance.

Social Music Review Platform
PHP, JavaScript, PostgreSQL, jQuery, AJAX

- Built a full-stack social music review platform inspired by Letterboxd, owning the entire frontend from information architecture to interaction design.
- Engineered asynchronous interactions (review posting, wishlist updates, activity feed) using AJAX to eliminate full-page reloads and improve responsiveness.
- Designed a normalized PostgreSQL schema for user relationships, reviews, and activity feeds, supporting efficient cross-user queries across interconnected social features.
- Implemented a social graph enabling user follows and cross-user interactions, supporting real-time feed generation and personalized content views.

Event Management System
Python, Django, PostgreSQL, JavaScript, HTML/CSS

- Built a full-stack event coordination platform for campus organizations with role-based access control spanning users, organizers, and admins.
- Designed and implemented a permission system enforcing granular access rules for event creation, editing, and registration workflows.
- Automated testing and deployment using GitHub Actions CI/CD pipeline, ensuring every pull request passed validation before production deployment.
- Modeled relational database schema supporting multi-organization event scheduling and user participation tracking.

Skills

Languages: Java, Python, JavaScript, C/C++
Backend & Frameworks: Spring Boot, Django, Node.js, Express, React
Databases: PostgreSQL, MongoDB
Cloud & DevOps: AWS (EC2, S3), Docker, GitHub Actions, Linux
Tools & Testing: Git/GitHub, JUnit, Apache Tomcat, Bruno